Mixed Hodge

① Hodge theory: X complex manifold. $\rightsquigarrow$ bicomplex

$$\begin{array}{ccc} \partial \uparrow & & \\ \Omega^{1,0} & \to & \Omega^{1,1} \\ \uparrow & & \uparrow \\ \Omega^{0,0} & \to & \Omega^{0,1} \xrightarrow{\bar\partial} \end{array}$$

$\Omega^{p,q}$ = p-hol q-anti-hol forms.

Dolbeault coh:

$$H^{p,q}(X;\mathbb{C}) = \frac{\ker(\bar\partial)}{\operatorname{im}(\bar\partial)}$$

Dolbeault's thm: $H^{p,q}(X;\mathbb{C}) \simeq H^q(X, \Omega^p)$ ↑ sheaf coh.

Hodge decomp: X compact Kähler (e.g. projective). Then

$$H^m(X;\mathbb{C}) \simeq \bigoplus_{p+q=m} H^{p,q}(X;\mathbb{C}).$$

Proof (for only. var.): ~~Lefschetz's theorem~~: $\Omega^{\bullet,\bullet} \to \underline{\mathbb{C}}_X$ qis of sheaves (Poincaré lemma) [analytic category]

- spectral seq deg at E_1 page $\searrow$ filter.

$$\begin{array}{ccccccc} 0 & \to & \Omega^0 & \to & \Omega^1 & \to & \cdots \\ & & & & \cup\!\shortmid & & \\ 0 & \to & \Omega^1 & \to & \cdots & & \\ & & \cup\!\shortmid & & & & \\ & & \vdots & & & & \end{array}$$

- Dolbeault's thm ⊛

$E_1^{p,q} = H^q(X, \Omega^p) \Rightarrow H^n(X;\mathbb{C})$.

weights: $H^0(X;\mathbb{C})$ pure of weight 0 (complex)

$H^n(X;\mathbb{C})$ pure of wt n

$H^n(X;\mathbb{C})[-n]$ pure of wt 0.

Hodge structure: ~~pq~~ $\mathbb{Z}$-module V with $V \otimes_{\mathbb{Z}} \mathbb{C} = \bigoplus_{p,q} V^{p,q}$.

weight $= p+q$.

Hodge number: $h^{p,q} = \dim(V^{p,q})$.

② Representation theory.

Def Tate: Hodge structure is Tate if $h^{p,q}=0$ when $p\neq q$.

Prop. $\tilde{\mathfrak{g}} \to \mathfrak{g}$ Cartan-Sp resolution. $\tilde{\mathfrak{g}} = \{(X,\mathfrak{b}) \in \mathfrak{g}\times G/B \mid X\in\mathfrak{b}\} \xhookrightarrow{i} \mathfrak{g}\times G/B \xrightarrow{\pi} G/B$, $\tilde{\mathfrak{g}} \mapsto \mathfrak{g}$

$H^i(\tilde{\mathfrak{g}}, \mathcal{O}_{\tilde{\mathfrak{g}}}) = 0$ when $i\neq 0$.

Proof: $H^i(\tilde{\mathfrak{g}}, \mathcal{O}_{\tilde{\mathfrak{g}}}) = H^i(G/B, R\pi_* i_* \mathcal{O}_{\tilde{\mathfrak{g}}}) = H^i(G/B, R\pi_* \mathcal{K}(\mathcal{O}_{\mathfrak{g}\times G/B})$

Koszul.

$$\mathcal{K}(\mathcal{O}_{\mathfrak{g}\times G/B}) = \left(\begin{array}{c} \mathcal{O}_{\mathfrak{g}\times G/B} \\ \uparrow \\ \mathcal{O}_{\mathfrak{g}\times G/B}\otimes \pi^*\Omega^1_{G/B} \\ \uparrow \\ \vdots \\ \uparrow \\ \mathcal{O}_{\mathfrak{g}\times G/B}\otimes \pi^*\Omega^n_{G/B} \end{array}\right)$$

proj. formula

$$\simeq H^i\left(G/B, \mathcal{O}(\mathfrak{g}) \otimes \left(\begin{array}{c} \mathcal{O}_{G/B} \\ \uparrow \\ \Omega^1_{G/B} \\ \uparrow \\ \vdots \\ \Omega^n_{G/B} \end{array}\right)\right)$$

sp. seq. $E_1^{p,q} = H^q(G/B, \Omega^p_{G/B}) \otimes \mathcal{O}(\mathfrak{g}) \Rightarrow H^{q-p}(\tilde{\mathfrak{g}}, \mathcal{O}_{\tilde{\mathfrak{g}}})$

good enough if $H^q(G/B, \Omega^p_{G/B}) \neq 0$ iff $q=p$ i.e. Tate.

Claim 1: $H^\bullet(G)$ Tate for any alg gp G.

Pf: $G = G'U$, but $U \simeq pt$. Assume G reductive.
rel. unip.

Splitting pr.: $H^\bullet(G) \hookrightarrow H^\bullet(T)$ Assume $G=T$ torus, esp. $G=\mathbb{G}_m$

$G=\mathbb{G}_m$: "allowed" Hodge #s: $h^{11}+h^{10}+h^{01}=1$, $h^{10}=h^{01}$ $\Rightarrow h^{11}=1$. Tate.

Why: need ① "nice" Hodge thy for nonsmooth
② know what Hodge #'s can appear in $H^k(X;\mathbb{Q})$.

Claim: $H^{\cdot}(X/G)$ Tate if $H^{\cdot}(X)$ Tate.

Pf: $H^{\cdot}(X/G) = H^{\cdot}\begin{pmatrix} \vdots \\ \text{sk}_i \\ \text{Cone} \\ H \\ X \end{pmatrix}$ each term is Tate.

Tolerantly is Tate.

Cor: $H^{\cdot}(G/B)$ is Tate.

③ mixed Hodge structure (abstract), $\boxed{\text{smooth, noncompact}}$

- $\mathbb{Z}$-mod V (think: $V = H^m(X;\mathbb{Z})$)
- incr. fil. W on $V \otimes \mathbb{Q} = V_{\mathbb{Q}}$
- decr. fil. H on $V \otimes \mathbb{C} = V_{\mathbb{C}}$

such that $gr_W^m(V_{\mathbb{C}})$ pure Hodge of weight m.

(specialize to Hodge str:
$F^{W,n} V = \bigoplus_{p+q \le n} V^{p,q}$. $F_{H,n} = \bigoplus_{p \ge n} V^{p,q}$)

$X \hookrightarrow M$ smooth. $M - X = D$ normal crossing divisor "good compactification"

Existence: Hironaka.

locally, in coords $(z_1, \dots, z_n)$, $D = z_1 \cdots z_k = 0$.
at $x \in M$

Def: $\Omega^0_{M,x}(\log D) = \Omega^0_{M,x} = \mathcal{O}_{X,x}$

$\Omega^1_{M,x}(\log D) = \mathcal{O}_{M,x} \frac{dz_1}{z_1} \oplus \cdots \oplus \mathcal{O}_{M,x} \frac{dz_k}{z_k} \oplus \mathcal{O}_{M,x} dz_{k+1} \oplus \cdots \oplus \mathcal{O}_{M,x} dz_n$

$\Omega^k_M(\log D) = \Lambda^k \Omega^1_M(\log D)$.

Thm: 1) $H^k(X;\mathbb{C}) \simeq \mathbb{H}^k(M, \Omega^\cdot_M(\log D))$.

2) weight filtr:

$$\begin{array}{ccccccc}
 & & \vdots & & \vdots & & \Omega^2_M(\log D) \wedge \Omega^1_M \\
 & & \uparrow & & \uparrow & & \uparrow \\
0 \subseteq & & \Omega^0_M(\log D) \wedge \Omega^1_M & \subseteq & \Omega^1_M(\log D) \wedge \Omega^1_M & \subseteq & \Omega^2_M(\log D) \\
 & & \uparrow & & \uparrow & & \uparrow \\
 & & \Omega^0_M(\log D) & & \Omega^1_M(\log D) & & \Omega^1_M(\log D) \\
 & & & & \uparrow & & \uparrow \\
 & & & & \Omega^0_M(\log D) & & \Omega^0_M(\log D)
\end{array} \subseteq \cdots$$

ie. weight k pure allows up to k $\frac{1}{z_i}$ (weight k = # of $\frac{1}{z_i}$ factors)

usual Hodge fil.

$\Rightarrow$ weight filtration is cohomology $\quad \text{image}\left(\mathbb{H}^k(M, W^{m-k}\Omega^\cdot_M(\log D))\right) \to H^k(X;\mathbb{C})$

$-k$ = "cohomology weight stuff"

3) independent of choice of good compactification.

Example: $X = \mathbb{A}^1 \subseteq \mathbb{P}^1 = M$

$dy = -y^2 dx$.

$$\begin{tikzcd}
\Omega^1_{\mathbb{P}^1}(\log \infty) \arrow[r] & k[x]dx \oplus k[y]\frac{dy}{y} \arrow[r] & \frac{k[x,y]}{xy=1}dx \\
\Omega^0_{\mathbb{P}^1}(\log\infty) \arrow[r] \arrow[u] & k[x]\oplus k[y] \arrow[r] \arrow[u] & \frac{k[x,y]}{xy=1} \arrow[u]
\end{tikzcd}$$

$H^2 = 0$: $\frac{dy}{y} \longmapsto -y\,dx$

$H^1 = 0$: $(x^m dx, y^{n-1}dy, x^m) \longmapsto x^m(1+y^{n+1})dx$

$H^0 = \mathbb{C}\cdot(1,1)$.

spectral seq.

$$\begin{array}{ccc}
0 & & 0 \\
\uparrow & & \uparrow \\
\mathbb{C} & \oplus & 0
\end{array}$$

Example: $X = \mathbb{G}_m \hookrightarrow \mathbb{P}^1 = M$.

$$\begin{tikzcd}
\Omega^1_{\mathbb{P}^1}(\log 0,\infty) \arrow[r] & k[x]\frac{1}{x}dx \oplus k[y]\frac{1}{y}dy \arrow[r] & \frac{k[x,y]}{xy}dx \\
\Omega^0_{\mathbb{P}^1}(\log 0,\infty) \arrow[r] \arrow[u] & k[x]\oplus k[y] \arrow[r] \arrow[u] & \frac{k[x,y]}{xy=1} \arrow[u]
\end{tikzcd}$$

$H^2 = 0$.

$H^1 = \mathbb{C}\cdot\left(\frac{dx}{x} + \frac{dy}{y}\right)$

$H^0 = \mathbb{C}\cdot(1,1)$

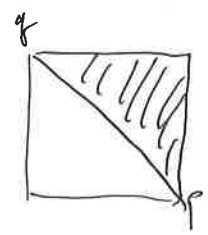
$$\begin{array}{ccc}
\frac{dx}{dx} + \frac{dy}{y} & \oplus & 0 \\
\uparrow 0 & & \uparrow \\
\mathbb{C} & \oplus & 0
\end{array}$$

"pre" weight $= 1$
coh shift $= 1$ $\rightsquigarrow$ weight $= 2$.
Hodge $= 1$.

$$\boxed{\begin{array}{c} h^{1,1} = 1 \\ h^{0,0} = 1 \end{array}}$$

Note: If $D = \emptyset$, weight filt trivial, everything is weight 0.

Prop: If X smooth, $H^k(X, \mathbb{C})$ has $h^{p,q} \neq 0$ only if $p \leq k$, $q \leq k$, $p+q \geq k$.



Pf: $p \leq k$: $H^i(\Omega^{>k})$ is in degree $> k$.

$q \leq k$: symmetry.

$p+q \leq k$: $p+q-k$ = "non-shifted weight", can't be negative.

Proof (Sketch) ① Show $\Omega^\cdot_M(\log D) \longrightarrow j_* \Omega^\cdot_X$ qis. to locally.

locally: $\Gamma(\mathcal{U}, \Omega^\cdot_M(\log D)) \longrightarrow \Gamma(\mathcal{U}, \Omega^\cdot_X)$

$$\begin{tikzcd} \Gamma(\mathcal{U}, \Omega^\cdot_M(\log D)) \arrow[r] \arrow[dr] & \Gamma(\mathcal{U}, \Omega^\cdot_X) \arrow[d, "\simeq"] \\ & H^\cdot(\mathcal{U}; \mathbb{C}) \end{tikzcd}$$

Poincaré Lemma.

~~dR theorem~~.

enough to show is iso. Note ~~$\mathcal{U} \simeq \prod_k S^1$~~ $\mathcal{U} \simeq \prod_k \mathbb{C}_m \simeq \prod_k S^1$

so $H^\cdot(\mathcal{U}; \mathbb{C}) \simeq \Lambda^\cdot \mathbb{C}\{\frac{dz_i}{z_i}\}_{i=1,\dots,k}$.

~~map~~ this defines map; to calculate (filter $\Omega^\cdot_M(\log D)$ by pole order)

② annoying; a lot of spectral sequences, filtrations, etc.

③ functoriality: take $X \hookrightarrow M$, $X \hookrightarrow N$

$$\begin{tikzcd} & Z \arrow[d] \\ X \arrow[ur, hook] \arrow[r, hook] & M \times N \end{tikzcd}$$

Z good resolution of closure

i.e.

$$\begin{tikzcd} X \arrow[r, hook] & Z \arrow[dl] \arrow[dr] & \\ M & & N \end{tikzcd}$$

$Z \to M$, $Z \to N$ birationally, so f^* is isom on $H^\cdot(X)$.

④ singular case: use proper hypercoverings.

<u>Idea</u>: $\mathbb{Q}_X$ sheaf on X nonsmooth: resolve $\mathbb{Q}_X$ with sums of sheaves $\mathbb{Q}_{X_i}$ where X_i smooth, (have Hodge/weights on $\mathbb{Q}_{X_i}$).

How to get such resolutions? "simplicial resolutions".

<u>Def</u>: <u>simplicial category</u> Obj: $[n]=\{0,1,2,\dots,n\}$. Mor: ~~nonc~~ nondecreasing maps

<u>n-simplical</u>: $n \le m$

<u>cubical category</u>: Obj: subsets of $\mathbb{N}$ Mor: $\subseteq$

<u>m-cubical</u>: subset size $|S| = m$

<u>simplical object</u>

$$\begin{array}{c} \vdots \\ X_2 \\ \downarrow\uparrow\downarrow \\ X_1 \\ \downarrow\uparrow\downarrow \\ X_0 \\ \downarrow \\ X \end{array}$$

augmentation

<u>semisimplical object</u>

$$\begin{array}{c} \vdots \\ X_2 \\ \downarrow\downarrow\downarrow \\ X_1 \\ \downarrow\downarrow \\ X_0 \\ \downarrow \\ X \end{array}$$

aug

<u>Example</u>: if $f: X \to Y$,

$$\begin{array}{c} X \times_Y X \times_Y X \\ \downarrow\downarrow\downarrow \\ X \times_Y X \\ \downarrow\downarrow \\ X \\ \downarrow \\ Y \end{array}$$

3-<u>cubical object</u>

$$\begin{tikzcd}
X_{\{0,1,2,3\}} \arrow[rr] \arrow[dr] \arrow[dd] & & X_{\{0,2,3\}} \arrow[dr] \arrow[dd] & \\
& X_{\{0,1,3\}} \arrow[rr] \arrow[dd] & & X_{\{0,3\}} \arrow[dd] \\
X_{\{1,2,3\}} \arrow[rr] \arrow[dr] & & X_{\{0,3\}} \arrow[dr] & \\
& X_{\{1,3\}} \arrow[rr] & & X_{\emptyset}.
\end{tikzcd}$$

<u>Ex</u>: if $X_0, X_1, X_2 \to Y$

$X_{\{0,1,2\}} = X_0 \times_Y X_1 \times_Y X_2$

$X_{\{1,2\}} = X_1 \times_Y X_2$

etc.

<u>Note</u>: cubical obj $\rightsquigarrow$ simplicial object.

$X_0 \longmapsto X_1 = \coprod_{|I|=m-1} X_I$.

<u>sheaf on simplicial space</u> $X_\bullet$: $\mathcal{F}_i$ on each X_i and maps $\mathcal{F}_i \to f_* \mathcal{F}_j$ (where it makes sense)

<u>gives</u>: (for $X_\bullet \xrightarrow{e} Y$),

full cohomologic double complex:

$$\begin{array}{c} \vdots \\ \downarrow \\ Re_{2*}\mathcal{F}_2 \\ \downarrow \\ Re_{1*}\mathcal{F}_1 \\ \downarrow \\ Re_{0*}\mathcal{F}_0 \end{array}$$

<u>Note</u>: Say $f: X \to Y$

$X_\bullet = (X \times_Y \cdots \times_Y X) \simeq \mathrm{Cech}(f)$

$\mathcal{F} \in Sh(Y) \rightsquigarrow \mathcal{F} \in Sh(X_\bullet)$

because f surjective,

$\mathcal{F} \to p_* p^* \mathcal{F}$.

<u>cohomological descent</u> ("resolution") $X_\bullet \to Y$ is of coh. descent if

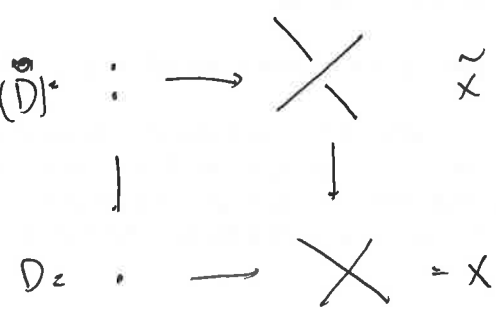
$\mathbb{Z}_Y \to e_* \mathbb{Z}_X$ is qis.

<u>Prop</u> $f: X_\bullet \to Y$ ~~is of coh. descent~~, f proper, with contractible fibers. Then f is of coh. descent.

<u>Def</u>: <u>proper modification</u> of X is $f: \tilde{X} \to X$, isom on dense open, <u>resolution</u> if $\tilde{X}$ smooth, <u>discriminant</u> is $X - (\text{open})$.

<u>Thm</u> (Abramovich, de Jong): X irred alg. variety, $D \subseteq X$ closed subset. Then $\exists$ resolution $f: \tilde{X} \to X$ where $f^{-1}(D)$ normal crossings divisor.

<u>Ex</u>: $f^{-1}(D)$:



$$\begin{array}{ccc} f^{-1}(D) & \longrightarrow & \tilde{X} \\ \downarrow & & \downarrow \\ D & \longrightarrow & X \end{array}$$

Note: square is st cohomological descent (gen'l. has contractible fibers)

Ex: $f^{-1}(D)$ ⋮ ↓

$\tilde{X} \amalg D =$ $\xrightarrow{BR}$

↓ ↓

$X =$

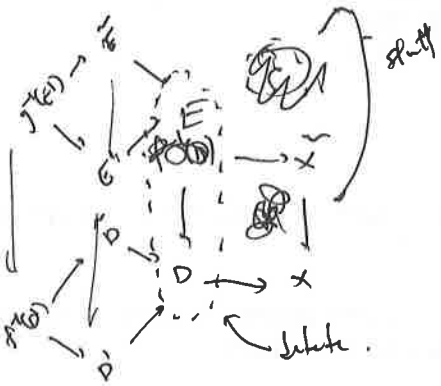
Problem: D, $f^{-1}(D)$ maybe not smooth.

Solution: induct. e.g. say we have

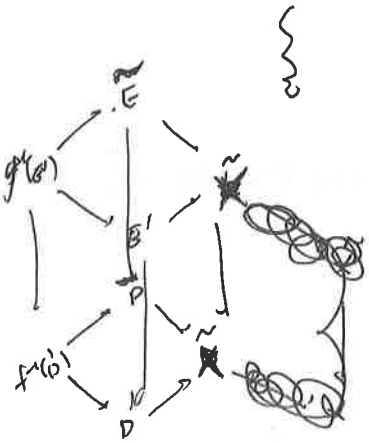
$$\begin{array}{ccc} f^{-1}(D) & \longrightarrow & \tilde{X} \\ \downarrow & & \downarrow \\ D & \longrightarrow & X \end{array}$$

↑ face where $S \ni k$. ← largest value.

Take a resolution of this $(k+1)$-cube to get $(k+3)$-cube. (incr. dim by 2)



Remove all vertices where $k \in S$ (i.e. delete a face, then use composition to fill in).



Cut $(k+2)$-cube. (increase dim by $2-1=1$)

Thm This eventually terminates

Pf: Each inductive step decreases dim of discriminant.

Hodge + weight structure: If we have qis:

$$\begin{tikzcd}
 & \vdots \arrow[d] & \vdots \arrow[d] & \vdots \arrow[d] & \vdots \\
 & \Phi_{X_1} \arrow[d] & R\varepsilon_*(\Omega^0_{X_1} \arrow[r] \arrow[d] & \Omega^1_{X_1} \arrow[r] \arrow[d] & \Omega^2_{X_1} \arrow[r] \arrow[d] & \cdots) \\
\Phi_X & \Phi_{X_0} \arrow[l, "\sim"'] \arrow[r, squiggly] & R\varepsilon_*(\Omega^0_{X_0} \arrow[r] & \Omega^1_{X_0} \arrow[r] & \Omega^2_{X_0} \arrow[r] & \cdots)
\end{tikzcd}$$

(flip Γ)

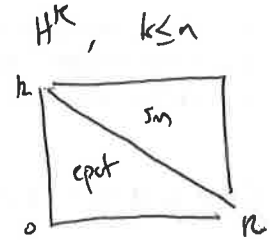
Hodge filt. is the usual, level-wise

weight filtration is shifted by $-(\text{row})$ i.e.

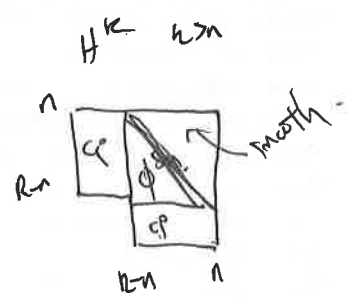
$$\Omega^0_{X_r} \to \Omega^1_{X_r} \to \Omega^2_{X_r} \to \cdots$$

has weights $r-q,\ r-q+1,\ r-q+2,\ \ldots$

Allowed Hodge numbers: $h^{p,q}$ in $H^k(X)$.

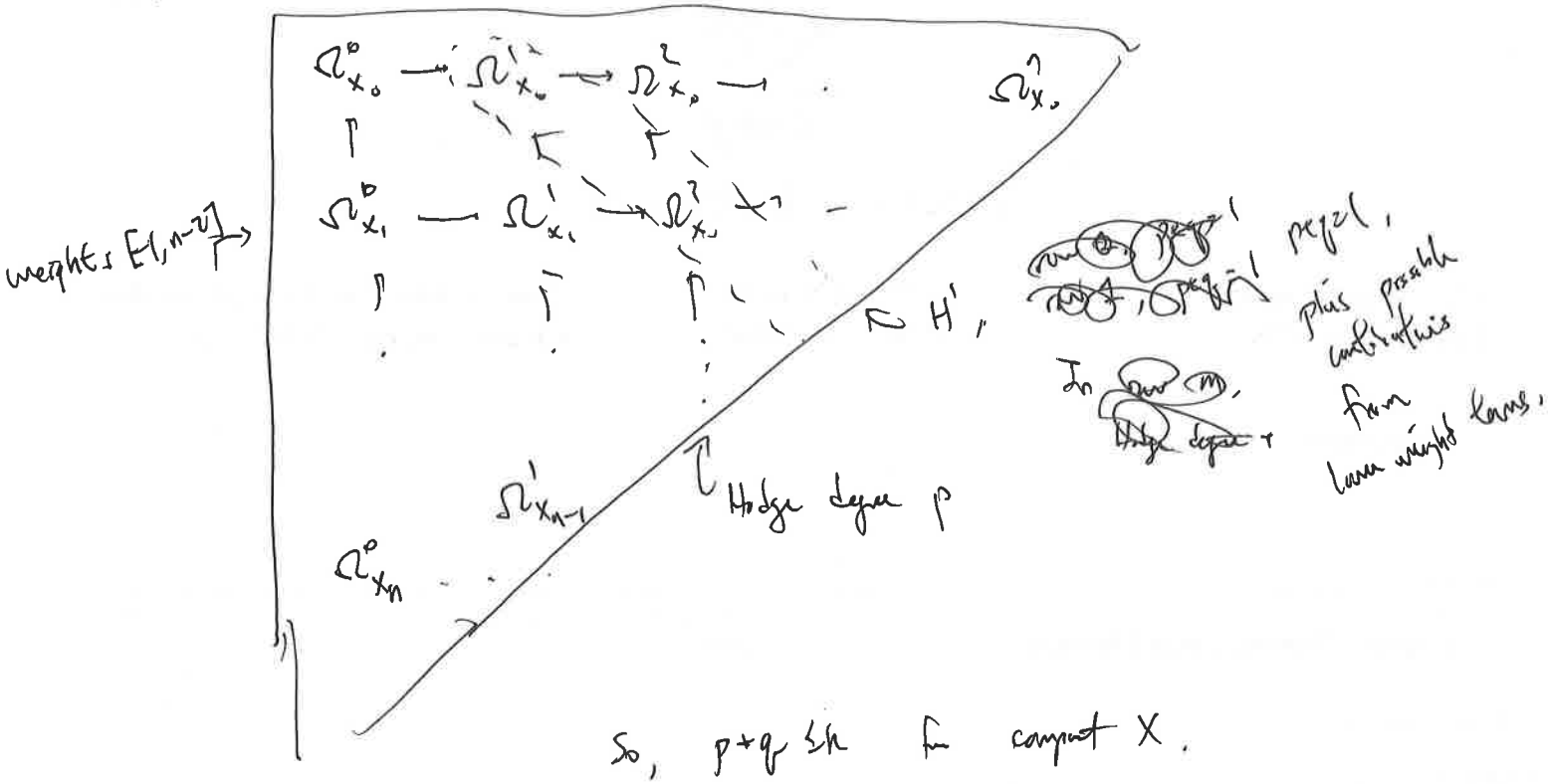


- $0 \le p, q \le k$
- If $k > n$, then $k-n \le p, q \le n$
- U smooth $\Rightarrow$ $p+q \ge k$
- U compact $\Rightarrow$ $p+q \le k$.



<u>Proof</u>: Because of $R\varepsilon_a$, can get smaller Hodge degree in H^k

i.e. If X compact, then



So, $p+q \leq k$ for compact X.

For $0 \leq p, q \leq k$, note that weights for $\searrow$ sing $= k$.

get contributions from possibly lower weights.

if $k > n$, then only contributions are from lower weights,

"cleared" by $R\varepsilon_a$, ...

<u>Mixed Hodge modules</u> X $\mathbb{C}$-alegb. variety, $\rightsquigarrow$ abelian category $MHM(X)$.

①
$$\begin{tikzcd} D^bMHM(X) \arrow[r, "rat"] \arrow[dr, "Dmod"'] & D^b_c(X,\mathbb{Q}) \arrow[r, "-\otimes\mathbb{C}"] \arrow[d, phantom, "\Rightarrow"] & D^b_c(X;\mathbb{C}) \\ & D^b_{Ch}(\mathcal{D}_X) \arrow[ur, "\simeq"'] & \end{tikzcd}$$

② $MHM(pt)$ = graded polarizable mixed Hodge structures

③ $M \in MHM(X)$ has <u>weight filtration</u>, graded pieces <u>semisimple</u>

④ Verdier duality functor

⑤ $f_!, f^*$ decreases weights

⑥ $f^!, f_*$ increases weights

⑦ if f proper $f_* = f_!$, preserves weights

⑧ if f smooth, $f^* = f^![n]$

f^* preserves weights.

<u>Ex</u>, $p: X \to pt$. $\mathbb{Q}_X = p^*\mathbb{Q}$

$H^\cdot(X,\mathbb{Q}) = p_*p^*\mathbb{Q}$

If X smooth, $\mathbb{Q}_X$ pure of zero, but. $p_*\mathbb{Q}_X = H^\cdot(X;\mathbb{Q})$ might have weights ≥ 0 (i.e. H^k weights $\geq k$.)
"preq"

X compact. $\mathbb{Q}_X$ weights ≤ 0, $p_*(p_! = p_!)\mathbb{Q}_X = H^\cdot(X;\mathbb{Q})$ has weights ≤ 0 (H^k weights $\leq k$).

<u>Note</u>: If X proper, "classical" Hodge filtration agree with above.